

```

$ cd /var/www/
$ sudo php -f hbridge.php
Enter choice:1
You pressed:1
Forward-1 motor on
Enter choice:1
You pressed:1
Forward-1 motor stop
Enter choice:2
You pressed:2
Reverse-1 motor on
Enter choice:2
You pressed:2
Reverse-1 motor stop
Enter choice:3
You pressed:3
Forward-2 motor on
Enter choice:3
You pressed:3
Forward-2 motor stop
Enter choice:4
You pressed:4
Reverse-2 motor on
Enter choice:4
You pressed:4
Reverse-2 motor stop
Enter choice:5
You pressed:5
All motor stop
Enter choice:6
You pressed:6
Not a valid command Enter choice:

```

Fig. 3: Program output

to the prompt. Once completed, install PHP-GPIO libraries to interact with the GPIO. Download libraries from <https://github.com/aaronpearce/PHP-GPIO>.

Extract libraries and paste them into '/var/www/'. (Take care that you paste only the files, not the folder itself.)

Download the source code (hbridge.php) and copy it into '/var/www/'. Now use 'cd' command to go to the directory where the source code is located and use the command below to run the software:

```
sudo php -f hbridge.php
```

where 'hbridge.php' is the name of the software.

You should get the output as shown in Fig. 3. You can now enter the choices to make the motors move as indicated in Table I.

Project 2: PWM generator

GPIO.php can do a lot of manipulation of GPIO pins of Raspberry Pi but beyond that it cannot do anything. Raspberry Pi is capable of handling many more functions like PWM, shift library, SPI, PC, LCD, transmit and receive, which are beyond the capability of GPIO.php.

Therefore let's turn to the WiringPi module of PHP, which is an Open

Source software. Gordon Henderson made WiringPi module first in 'C,' which was then ported to Python, Ruby and PHP. Wiring is the core software that is used to control the various ports of Arduino development board. WiringPi is a similar package that works on Raspberry Pi. Commands are similar as in Arduino, and various sensors, when connected to Raspberry Pi, communicate the same way.

In this project we will use WiringPi-PHP and program in Arduino-like instructions to generate PWM signal at GPIO1 (physical pin 12). Installation of WiringPi-PHP is a bit tricky. Below-mentioned steps have to be carefully followed to successfully do it.

WiringPi-PHP installation

First, do an update using the command:

```
sudo apt-get update
```

You need to install php5-dev before installing WiringPi-PHP. The command for installation is:

```
sudo apt-get install php5-dev
```

Now install 'git' to install WiringPi-PHP:

```
sudo apt-get install git
```

Clone the WiringPi-PHP repository using:

```
git clone https://github.com/WiringPi/WiringPi-PHP.git
```

Now using 'cd' command go to the downloaded directory:

```
cd WiringPi-PHP
```

Now write 'git submodule update -init' command and press 'Enter.' Finally, build and install WiringPi-PHP using commands:

```
sudo ./build.sh
sudo ./install.sh
```

Copy wiringpi.php to '/var/www/' location using command:

```
cp wiringpi.php /var/www/
```

Now wiringpi is installed on your Raspberry Pi and also copied to the '/var/www' directory. But to get it into PHP, you need to load it in '.ini' file using following steps:

1. Open the wiringpi.ini file at '/etc/php5/conf.d/' location using



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